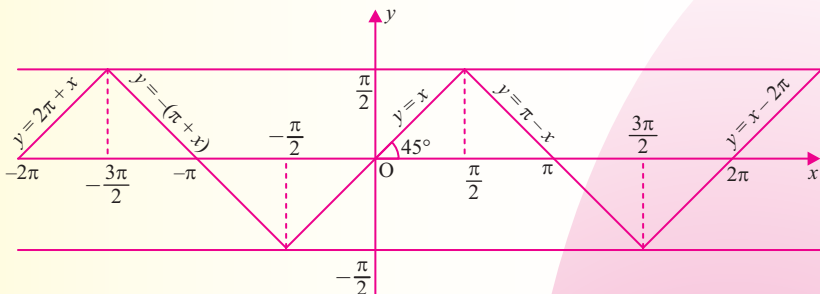
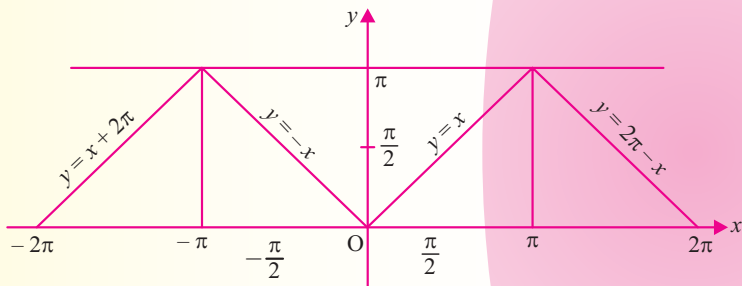


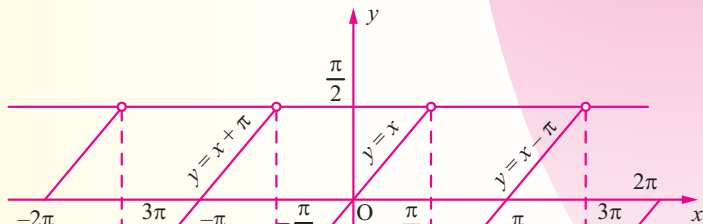
[2 2]



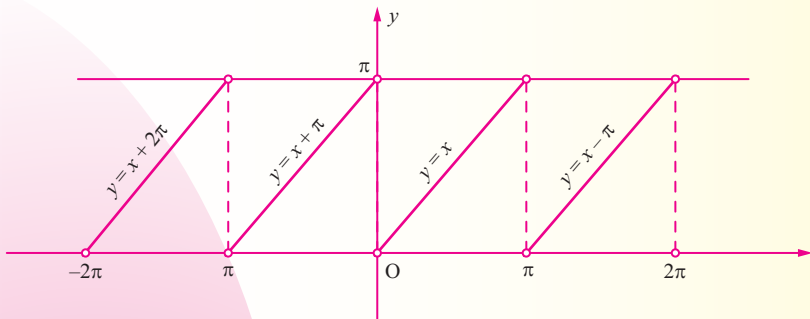
(ii) $y = \cos^{-1}(\cos x)$, $x \in R$, $y \in [0, \pi]$, periodic with period 2π .



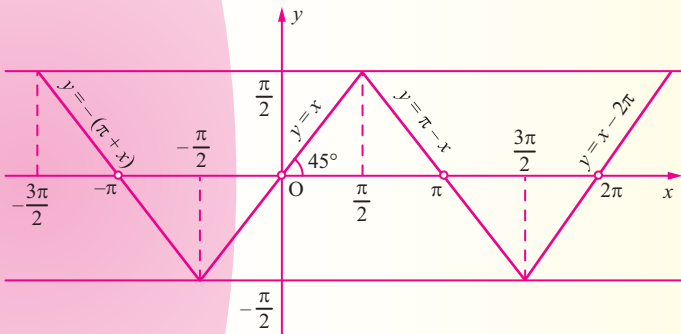
(iii) $y = \tan^{-1}(\tan x)$

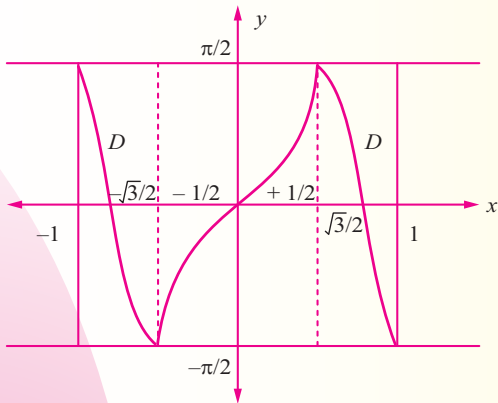


(iv) $y = \cot^{-1}(\cot x)$, $x \in \mathbb{R} - \{n\pi\}$, $y \in (0, \pi)$, periodic with period π .



(v) $y = \operatorname{cosec}^{-1}(\operatorname{cosec} x)$, $x \in \mathbb{R} - \{n\pi, n \in \mathbb{I}\}$, $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$, periodic with period 2π .

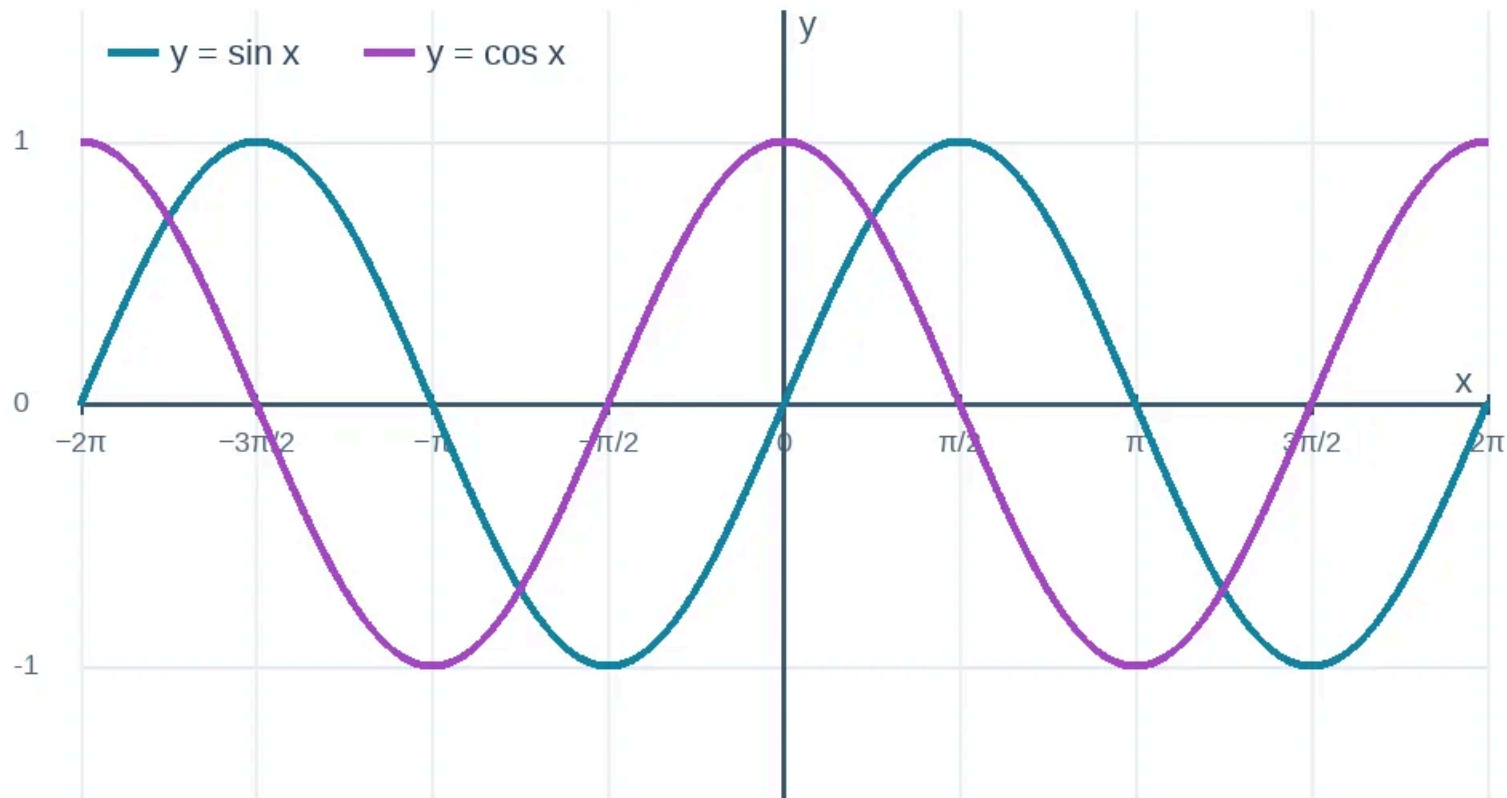




$$(e) \ y = f(x) = \cos^{-1}(4x^3 - 3x) = \begin{cases} 3 \cos^{-1} x - 2\pi & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 2\pi - 3 \cos^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3 \cos^{-1} x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

y

Trigonometric function graphs



Tangent function graph

